Chronic Kidney Disease: Exploratory Data Analysis to unveiling clinical patterns on R language

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Background

Chronic kidney disease (CKD)

- Abnormalities of kidney structure or function (≥ 3 months)
- Global health burden: **700 million** people affected
- **Asymptomatic** early stages

→ *We analyze a dataset to obtain statistical insights for CKD.*

Dataset

- Open accessed; Clinical measurements; From India
- **Raw:** 400 samples & 25 features
- **Cleaned:** 369 samples & 28 features & 0 missing values

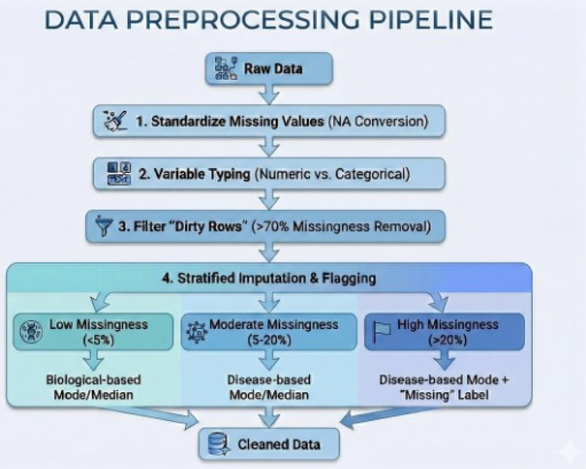


Fig. 1. The full pipeline of dataset preprocessing in R

Research Questions

1. **Clinical Baselines:** What is the distributional characteristics of **Age (age)** and **CKD Diagnosis (classification)**?
2. **Pathological Characteristics:** Is there a difference between means of **Hemoglobin levels (hemo)** between CKD and non-CKD individuals?
3. **Impact of Complications:** Do **Hypertension (htn)** and **Diabetes Mellitus (dm)** exhibit an interactive effect on kidney function as measured by **Serum Creatinine (sc)**?
4. **Metabolic Correlation:** To what extent does **Blood Urea (bu)** linearly predict **Serum Creatinine (sc)** concentration?

Methods

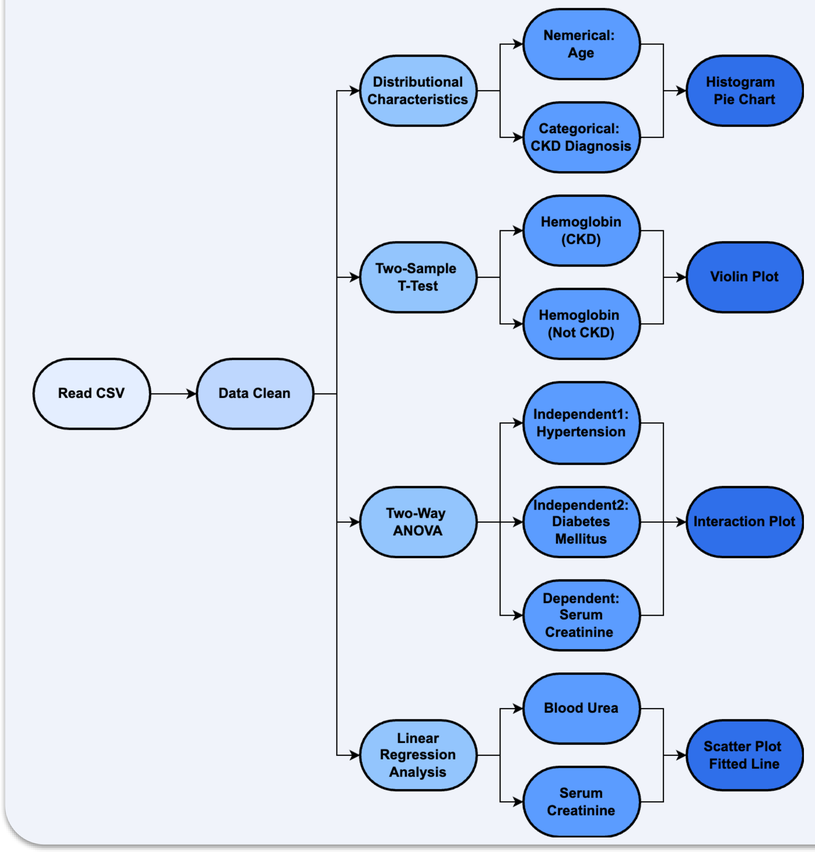


Fig. 2. Workflow of the whole research

Results & Discussion

Research Question 1 (Distributional Characteristics):

- **Age (numerical):**
  - Mean: 51.48 (years) | Median: 54 (years) | Mode: 60 (years) → Mainly middle-aged patient
  - SD: 16.72 (years) | Range: 90 – 3 = 87 (years)
  - Skewness: -0.65 | (Excess) Kurtosis: 0.072 → Negative skewness & mesokurtic
- **CKD Diagnosis (categorical):**
  - 219 (59.3%) CKD VS. 150 (40.7%) Non-CKD



Fig. 3. Proportion of CKD Diagnosis



Fig. 4. Distribution of Age

Research Question 2 (Two-Sample T-Test):

- **Pre-evaluation:**
  - Independence + Normality (Sample Size/Shapiro-Wilk Test/QQ Plot) + Variance Homogeneity (Levene's Test)
- **Two-Sample T-Test:**
  - $t_{(df = 362.72)} = 25.99$ + 95 CI: [4.14, 4.81] (g/dL)
  - p-value < 2.2e-16 → reject H_0
- **Conclusion:**
  - Significant difference between means exists

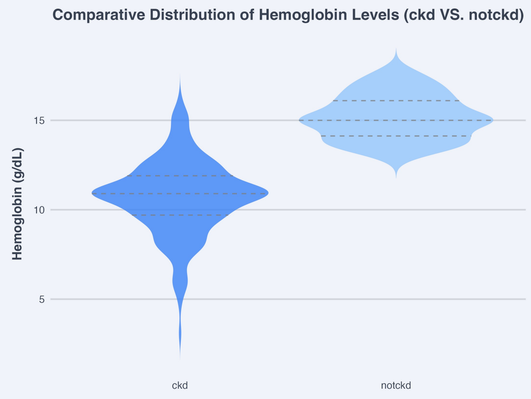


Fig. 5. Violin plot of hemoglobin levels

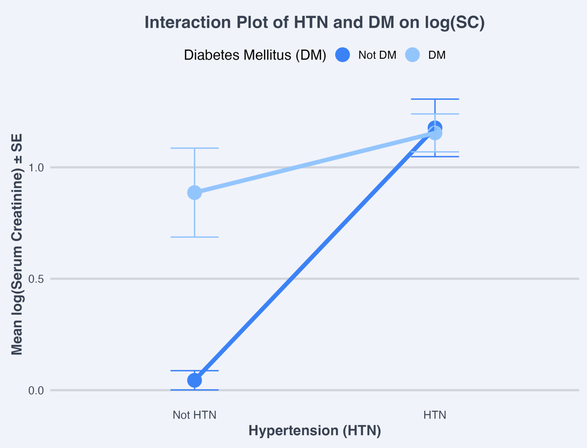


Fig. 6. Interaction plot of hypertension & diabetes mellitus effect on serum creatinine

Research Question 3 (Two-Way ANOVA):

- **Pre-evaluation:**
  - Independence + Normality (Log) + Variance Homogeneity (un-equal, use `Anova()` for correction)
- **Two-Way ANOVA:** ($F_{(df1 = 1, df2 = 365)}$)
  - HTN: F = 67.44 & p-value = 3.8e-15 → reject H_0
  - DM:F = 16.34 & p-value = 6.4e-05 → reject H_0
  - (HTN:DM): F = 11.02 & p-value = 0.0010 → reject H_0
- **Conclusion:**
  - Both HTN & DM have significant main effects on SC
  - No effect of DM in Not HTN - Effect of DM in HTN
  - Effect of (DM*HTN) Interaction (positive)

Research Question 3 (Linear Regression):

- **Linear Regression Analysis:**
  - Equation: $y = 0.060x - 0.66$
  - Adjusted $R^2 = 0.63$
  - p-value < 2.2e-16
- **Conclusion:**
  - A relatively strong positive linear relationship
  - 63% of the variation SC can be explained by BU

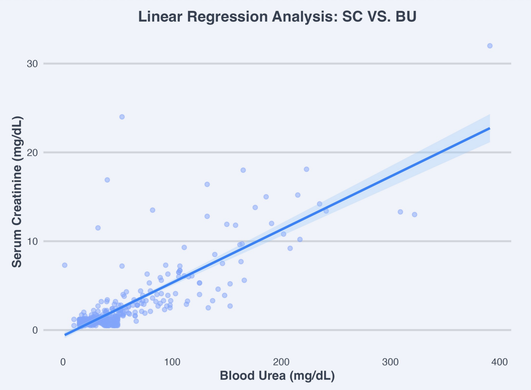


Fig. 7. Linear Regression for SC vs. BU

Conclusion

1. Middle-aged prevalence→ Early screening & intervention
2. Hemoglobin reduction in patients → Diagnostic marker
3. Hypertension & Diabetes → Synergistic complication
4. linear predictivity (Blood Urea)→ Easier monitoring

- **Limitations:**
  - Single geographic demographic
  - Simple co-factor analyses
- **Future Direction:**
  - Machine Learning → CKD prediction

Reflection & AI Usage

- **AI Usage** (ChatGPT)
  - Figure/Code/Language optimization
  - All ideas are NOT from AI
  - All AI generated context are double checked

References

- *Francis, A., Harhay, M. N., Ong, A. C. M., Tummalapalli, S. L., Ortiz, A., Fogo, A. B., Fliser, D., Roy-Chaudhury, P., ... & Jha, V. (2024). Chronic kidney disease and the global public health agenda: an international consensus. Nature Reviews Nephrology, 20, 473–485. https://doi.org/10.1038/s41581-024-00820-6*
- *Rubini, L., Soundarapandian, P., & Eswaran, P. (2015). Chronic Kidney Disease [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5G020*